Automatic Modulation Classification Based on Constellation Density Using Deep Learning

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***Abstract*— Deep learning (DL) is a newly addressed area of research in the field of modulation classification. In this letter, a constellation density matrix (CDM) based modulation classification algorithm is proposed to identify different orders of ASK, PSK, and QAM. CDM is formed through local density distribution of the signal's constellation generated using LabVIEW for a wide range of SNR. Two DL models, ResNet-50 and Inception ResNet V2 are trained through color images formed by filtering the CDM. Classification accuracy achieved demonstrates better performance compared to many existing classifiers in the literature.**

***Index Terms*— Modulation classification, deep learning, constellation, color image.**

I. Introduction

AUTOMATIC Modulation Classification (AMC) is a process to identify the modulation scheme of received unknown Radio Frequency (RF) signal. It is an intermediate step between signal interception and demodulation. AMC is broadly divided into two categories: Likelihood Based (LB) and Feature-Based (FB). LB modulation classifier compares the likelihood function value of the received signal within the modulation pool considered [1]. It has also been used in a multipath channel environment for estimation of unknown parameters and provides prominent results for modulation classification [2]. The LB methods need channel parameters to be known and become computationally complex when unknown parameters are introduced. Hence FB methods are developed, which are sub-optimal with lower computation complexity.

In FB methods, features of the received signal are extracted, and the modulation scheme is identified either by comparing these with threshold values or by feeding features to a pattern recognizer. Features like moments, cumulants, higher-order statistics, and spectral based are the most frequently used features in literature [1], [3]. Fold-based Kolmogorov–Smirnov method has also been applied with necessary features for efficient modulation classification [4]. Machine Learning (ML) is another advanced technique, which includes different classifiers like Artificial Neural Network (ANN), K-Nearest Neighbor (KNN), stacked autoencoders, Genetic Programming-KNN (GP-KNN) [5], etc. Clustering-based methods have also been applied to constellation for modulation classification [6]. ML-based methods provide adequate classification accuracy, but features need to be manually selected, which is overcome by Deep Learning (DL) methods. In [7], the color image is generated using constellation density and further used for classification using AlexNet and GoogleNet classifiers. Moreover, in [8] nonnegative constraint algorithm is used with DL autoencoder network for modulation classification.

This letter addresses modulation classification method using two DL models i.e. ResNet-50 [9] and Inception ResNet V2 [10]. The contributions of this letter are as follows: (1) Constellation Density Matrix (CDM) has been generated for the extraction of local information of the constellation. (2) Masking filters are designed to remove information corrupted due to high noise. Each filter has three masks to generate the color image, and each of the masks is designed to extract the information of one class. (3) Selected classification models learn high dimensional features and provide better classification results than other Convolutional Neural Networks (CNN) based models. The rest of the letter is organized into four sections. Section II describes the signal model. DL models architecture for classification and data pre-processing have been explained in sections III and IV, respectively. Results of the developed method have been given in section V.

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II. Signal Model

Signals modulated with M-Amplitude Shift Keying (MASK), M-Phase Shift Keying (MPSK), and M-Quadrature Amplitude Modulation (MQAM) are considered to be transmitted through Additive White Gaussian Noise (AWGN) channel. The received signal can be expressed as

$$y[n] = x[n] + \eta[n], \quad n = 1, 2, \ldots, N$$

where $y[n]$ is received complex symbol, $x[n]$ is transmitted complex symbol from one of the considered modulation scheme and $\eta[n]$ is AWGN of zero mean and σ_η^2 variance. Symbols are extracted with equal probability from the considered pool of modulation schemes. SNR for symbols from unit power constellation is defined as

$$SNR = 10log_{10}(\frac{1}{\sigma_\eta^2})$$

III. Deep Learning for Modulation Classification

Deep Networks consist of multiple layers with multiple neurons in each layer for the extraction of features autonomously. Initial layers extract abstract features, and deep layers get significant features by multiple nonlinear transformations on the previous layer's output. DL models extract these features without manual intervention, and it is called data-driven feature extraction.

A. ResNet-50

ResNet-50 is well known DL model in the field of image classification. Deep networks provide good classification results in comparison to shallower, but these are difficult

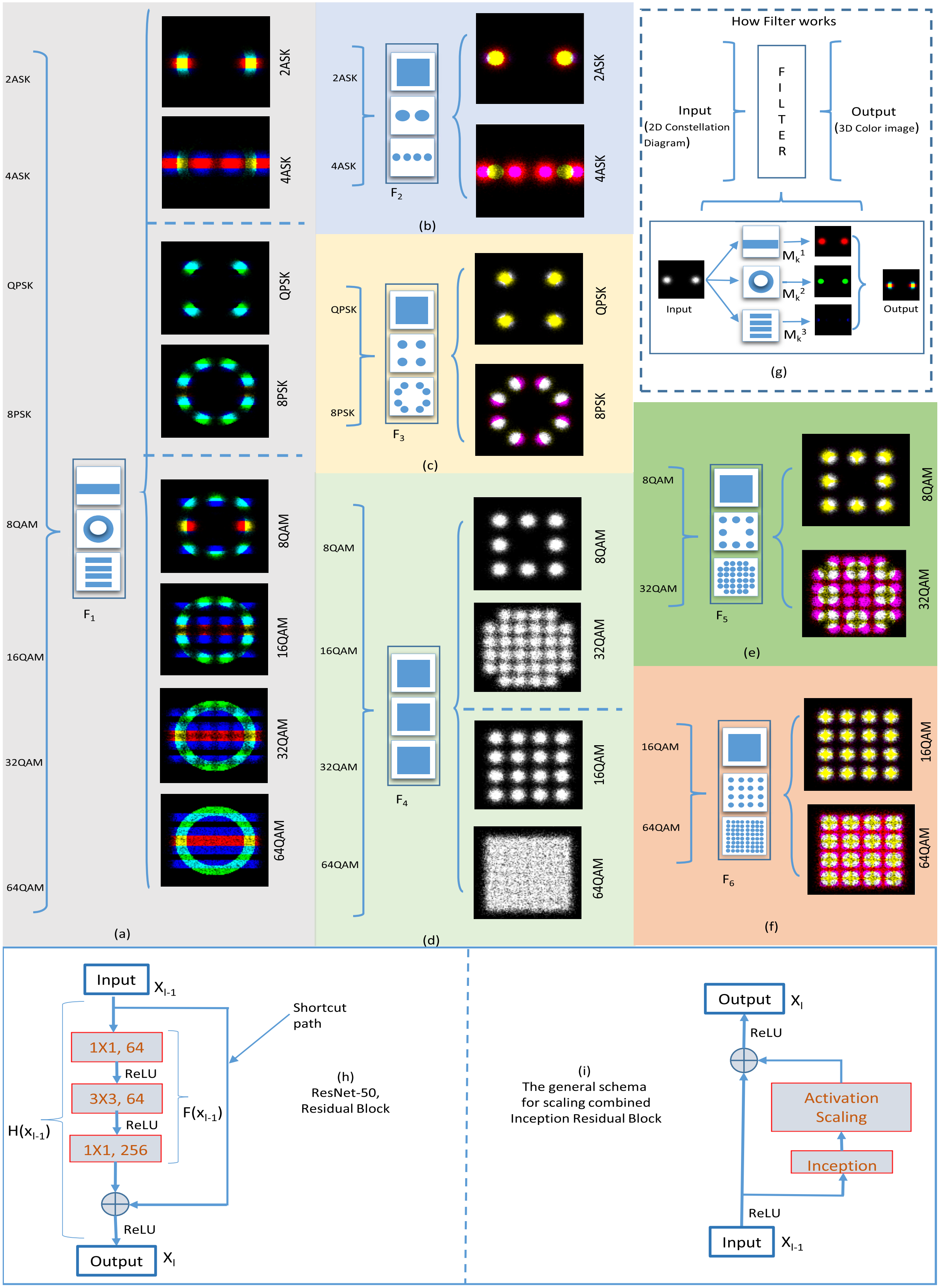

Fig. 1. A three-stage modulation classification procedure for eight modulation schemes with masking filters is shown. First stage: (a) constellation of all modulation schemes are filtered through F_1 and classified into three groups. Second stage: (b) constellation of 2ASK and 4ASK are filtered through F_2 and classified, (c) constellation of QPSK and 8PSK are filtered through F_3 and classified, (d) constellation of 8QAM, 16QAM, 32QAM and 64QAM are filtered through F_4 and classified into two groups. Third stage: (e) constellation of 8QAM and 16QAM are filtered through F_5 and classified, (f) constellation of 32QAM and 64QAM are filtered through F_6 and classified. (g) Description of the working procedure of all five filters. Functional architecture of (h) ResNet-50 Residual Block, and (i) Inception Residual Block.

to train due to vanishing gradient problem. This problem has been resolved by incorporating residual function [9] and three-layered bottleneck residual architecture, as shown in Fig. 1(h). For the given input x_{l-1}, mapping of a function $x_l = H(x_{l-1})$ is done using function $F(x_{l-1}) = H(x_{l-1}) - x_{l-1}$, where $F(x)$ is residual function. The bottleneck structure of residual architecture effectively reduced the computational complexity of the network, and the shortcut path eliminates the vanishing gradient problem. The pooling of features is used to reduce the distortion effect along with computational complexity. For this purpose mainly two types of pooling functions have been used. After the first convolution layer, max-pooling is done to extract initial low-level features, and after the last convolution operation, average pooling is used as all the final extracted features are equally important for decision making. Rectified Linear Unit (ReLU) activation function: $r(x) = max(0, x)$ has been used instead of traditional, for fast training process. In addition, the regularization method called $dropout$ has also been applied with 0.5 probability to eliminate overfitting for training data.

B. Inception ResNet V2

Inception ResNet V2 network is an extensive and efficient network in image classification. The configuration of $Inception\ Residual\ Block$ is shown in Fig. 1(i). Where inception block is a sub-network of traditional parallel-connected convolution layers, followed by batch normalization and activation layers. The computation complexity has been significantly reduced by introducing 1×1 filters at the earlier layers, and then larger filters in $Inception\ Block$ [10]. For residual summation, $Activation\ Scaling$ is required to compensate for the dimension reduction done by $Inception\ Block$. This model has parallel branches, which are easy to get trained on multiple GPUs. Very deep and wide inception network is implemented with intermediate auxiliary classifiers to reduce vanishing gradient problem while back-propagation. For neuron activation, the ReLU function is used, and for regularization, $dropout$ with a probability of 0.5 is used similar to ResNet-50.

IV. Data Pre-Processing and Training

This research aims to identify the correct modulation scheme among 2ASK, 4ASK, QPSK, 8PSK, 8QAM, 16QAM, 32QAM, and 64QAM. The constellation points are used to generate CDM and transformed into a color image with three channels generated by masking the CDM with a proper filter. Generated color images are used to train multiple DL models in three stages of the classification hierarchy.

A. Color Image Formation Using CDM

The signal has been transformed into a color image of size $3 \times 100 \times 100$, to facilitate both the considered models with striking features. After normalizing the constellation points with average energy, points in 4×4 area of the complex plane are considered for further processing. This range covers most of the points for SNR under consideration. Constellation is divided into 100×100 equal grid sections. The number of points in each section is calculated and scaled in (0,255) range. The 100×100 dimension matrix formed is CDM.

TABLE I
Filters Specifications

	R Channel	G Channel	B Channel
Filter1	Rectangle: (11X100)	Disc: R1=27; R2=37	4 Rectangle: (9X100)
Filter2	Square: (100X100)	2 Circles: R=8	4 Circles: R=6
Filter3	Square: (100X100)	4 Circles: R=18	8 Circles: R=10
Filter4	Square: (100X100)	Square: (100X100)	Square: (100X100)
Filter5	Square: (100X100)	8 Circles: R=10	16 Circles: R=6
Filter6	Square: (100X100)	32 Circles: R=8	64 Circles: R=4

To convert CDM into a color image, filter F_k is applied, where $k \in \{1, 2, \ldots, 6\}$. F_k consists of three masks M_k^1, M_k^2 and M_k^3, as shown in Fig. 1(g). Each mask is a Boolean matrix of dimension 100×100 with distribution according to Table I. It has Boolean '1' at different shapes provided, and the remaining elements of the matrix are valued Boolean '0'. The output of each mask represents one channel of RGB image. Output of m^{th} mask of F_k filter is given by

$$CDM(i,j) \cdot M_k^m = \begin{cases} CDM(i,j); & M_k^m = 1 \\ 0; & M_k^m = 0 \end{cases}$$

B. Classification Strategy

The hierarchy of three stages is used to classify a modulation scheme. In the first stage, the domain of modulation is identified in two steps. (1) CDMs of all modulation schemes for a range of SNR are filtered through F_1 to get color images. F_1 has a set of three masks, the first mask has a shape of a rectangular strip that covers all ASK constellations, second has a shape of a circular strip to cover PSK constellation, and third has four rectangle strips to cover the constellation of QAM. Color images generated from CDMs of all modulation schemes filtered through F_1 are shown in Fig. 1(a). (2) These images are given as an input to the DL model for three-class classification. In the second stage of classification hierarchy, three filters F_2, F_3 and F_4 are used to generate color images. If decision of first stage is in favour of ASK domain, filter F_2 is used to filter CDMs of 2ASK and 4ASK constellations, for PSK domain F_3 is used to filter CDMs of QPSK and 8PSK constellations, and for QAM domain F_4 is used to filter CDMs of 8QAM, 16QAM, 32QAM and 64QAM constellations as shown in Fig. 1(b), 1(c) and 1(d) respectively. In the third stage, 8QAM and 32QAM are classified with a binary classifier and remaining 16QAM and 64QAM with another binary classifier, where these classifiers are trained with color images generated through F_5 and F_6 as shown in Fig. 1(e) and 1(f) respectively. The description of dimensions used for all six filters is given in Table I, where each dimension depicts one grid section.

C. Model Implementation and Training

Pre-trained models of both ResNet-50 and Inception ResNet V2 architectures take an input image of size $3 \times 100 \times 100$ and classify for 1000 classes. In this work, binary and three-class classification task is performed by concatenating 8 fully connected layers at the end of each model. Starting 7 layers out of these 8 layers contains 1000, 750, 500, 250, 100, 50, 20 neurons, and the last layer contains 2 or 3 neurons according to the selected classifier. For training, -4 dB to

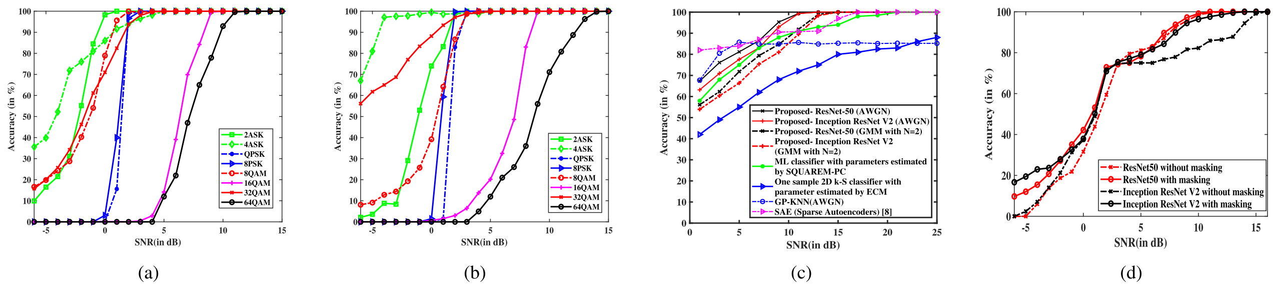

Fig. 2. Accuracy of developed method with (a) ResNet-50 and (b) Inception ResNet V2 for eight modulation schemes. (c) Proposed method comparison with GP-KNN in AWGN, ML classifier with parameter estimated by SQUAREM-PC in GMM (N = 2) and one sample 2D K-S classifier with parameter estimated by ECM in GMM (N = 2) for 2ASK, QPSK, 8PSK and 16QAM modulation schemes. (d) Comparison of average classification accuracy with and without masking filters for ResNet50 and Inception ResNet V2.

30 dB SNR with the step of 1 dB and 35 dB to 70 dB SNR with the step of 5 dB is considered, and 50 images at each SNR (2150 images in total) have been generated for each modulation scheme. In the first step of classification, the considered eight modulation schemes are divided into three groups, and 17200 (8×2150) images are given for training. In the second step, 2ASK and 4ASK have been classified with the model trained with 4300 images. Similarly, the hierarchy has been followed for the final classification. Both models are implemented in python with Keras libraries, and NVIDIA DGX-2 GPU has been used to train the models. Parameters of the solver configuration have also been adjusted for better classification and fast training processes, such as the learning rate is 0.0001, and the mini-batch size is 64. Batch size is considered based on the computation efficiency of GPU.

V. Results and Discussion

In this section, the performance of ResNet-50 and Inception ResNet V2 based methods for classification of eight modulation schemes are shown. The classification has been done in three stages, and probabilities of correct classification in consecutive stages are multiplied for final probability.

Modulation classification accuracy for considered modulation schemes is shown in Fig. 2(a) and 2(b) for ResNet-50 and Inception ResNet V2 models respectively. Results are obtained for 1000 signal realizations, each signal containing 50k symbol points. Both the models, ResNet-50 and Inception ResNet V2 provide reliable classification above 5 dB SNR for all modulation schemes except 16QAM and 64QAM. In comparison to all other modulation schemes, the classification accuracy of 16QAM and 64QAM is less, as both get confused with each other in their final stage of classification.

In Fig. 2(c), classification accuracy comparison of the proposed method with GP-KNN in the AWGN environment is given for BPSK, QPSK, 8PSK, and 16QAM modulation schemes. The proposed method achieves greater accuracy than GP-KNN for SNR higher than 7 dB. The proposed method is also compared with an ML classifier and one sample 2D K-S classifier in Gaussian Mixed Model (GMM) with N = 2, given in [11]. The proposed method with ResNet50 and Inception ResNet V2 models perform better than ML classifiers for 7 dB and above SNR. Also, the proposed method works better than one sample 2D K-S classifier for all values of SNR. All comparisons are done for 512 symbols.

To show the effect of masking filters, results without masking filters are also generated using the same hierarchy and channel condition for comparison as shown in Fig. 2(d). Performance gain in accuracy is achieved for both the models. Reasons for accuracy improvement is discarding the noisy constellation points by masking filters, and color image generation with three masks, each containing information of one class.

VI. Conclusion

In this letter, modulation classifiers are developed based on ResNet-50 and Inception ResNet V2 deep learning model using transfer learning. Constellation density of the received signals has been passed through some filters consisting of three masks to generate color images. These filters contain the inherent properties of the target classes in the spatial view. Modulation classification has been done in three stages by training both the models with generated color images. Comparison results show better performance of the proposed method than most of the methods available in the literature.

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